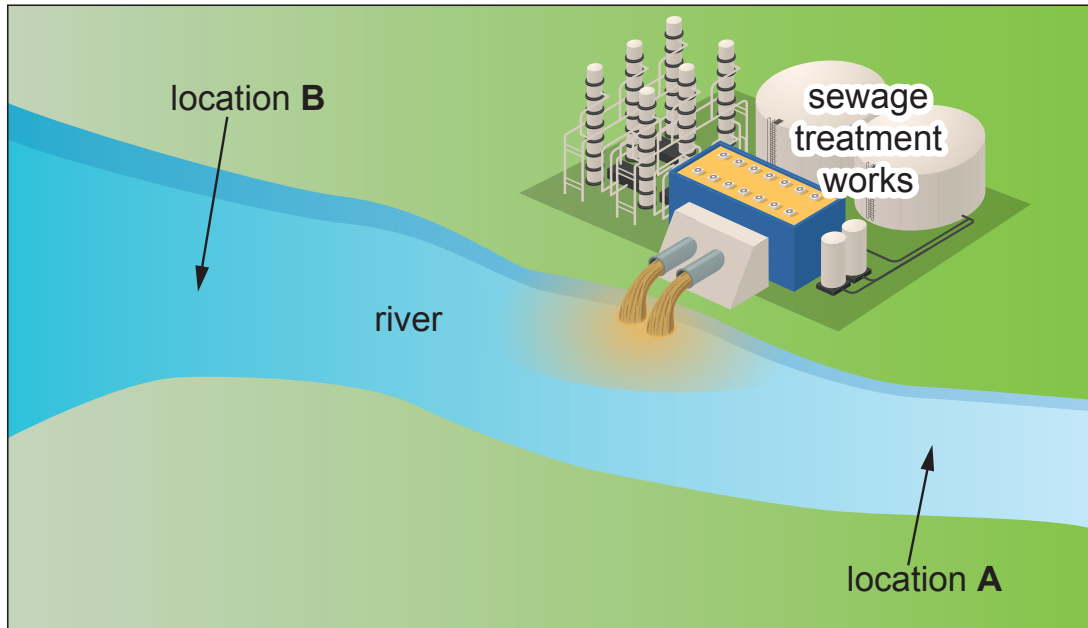


4. In 2022, a report by the Senedd Climate Change Committee noted that during 2020, there were over 105 000 incidents in Wales of untreated sewage being released into streams, rivers and the sea. Following extreme rainfall, sewage treatment works are allowed to release untreated sewage into streams, rivers and the sea.

Map 4.1 shows a river and the location of a sewage treatment works.

Map 4.1



In June 2020, a survey was carried out in a river at two locations, **A** and **B**, as shown on **Map 4.1**. The number and different types of invertebrates at each location was recorded. **Table 4.2** shows the results of the survey.

Table 4.2

Invertebrates	Number of invertebrates found	
	Location A	Location B
stonefly nymph	55	0
mayfly nymph	12	0
caddisfly larva	2	0
leech	3	0
sludge worm	0	33
rat-tailed maggot	0	27



Table 4.3 can be used to assess the level of pollution and oxygen concentrations in rivers.

Table 4.3

Invertebrates present	Level of pollution	Oxygen concentration
none	very high	no oxygen
rat-tailed maggot	high	low
sludge worm		
leech	moderate	moderate
blood worm		
caddisfly larva	low	moderate to high
mayfly nymph	none	high
stonefly nymph		

- (a) State the scientific term that is used to describe any organisms that can be used to monitor levels of pollution. [1]

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- (b) Use **Tables 4.2 and 4.3** to draw conclusions about the **oxygen concentrations** in the river at locations **A** and **B**. [2]

Oxygen concentration at location **A** =

Oxygen concentration at location **B** =

- (c) The release of untreated sewage into streams can result in an algal bloom, which in turn can reduce oxygen levels. Explain how algal blooms can result in reduced oxygen levels. [4]

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